

# What $M^{(L)}$ is: the length-filtration matrix of $\langle p_\rho, h_\nu \rangle$

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## Abstract

Let  $M$  be the matrix  $M_{\rho\nu} = \langle p_\rho, h_\nu \rangle = [m_\nu]p_\rho$  on partitions of  $n$ , and let  $M^{(L)}$  be its leading block on  $\{\rho : \ell(\rho) \leq L\}$ . In earlier work I proved that  $M$  is lower block-triangular for length, so that every  $M^{(L)}$  is invertible; the question left open was what  $M^{(L)}$  actually *is*. The answer is that the filtration is much finer than block-triangular: **each diagonal block is a diagonal matrix**, with entry  $\prod_i m_i(\rho)!$  at  $\rho$ . Consequently

$$\det M^{(L)} = \prod_{\substack{\rho \vdash n \\ \ell(\rho) \leq L}} \prod_{i \geq 1} m_i(\rho)! = \prod_{\substack{\rho \vdash n \\ \ell(\rho) \leq L}} \frac{z_\rho}{\rho_1 \cdots \rho_{\ell(\rho)}},$$

which is 1 exactly when  $L = 1$ , or  $n = 1$ , or  $L = 2$  with  $n$  odd. The inverse is the Möbius function of the partition lattice, it respects the same filtration, and the row of  $M^{-1}$  indexed by  $\nu$  has denominator exactly  $\prod_k m_k(\nu)!$  — so the  $h$ -coefficients are recovered from the  $p$ -coefficients by dividing by automorphism orders and nothing else. I give  $(M^{(3)})^{-1}$  explicitly and deduce the  $L = 3$  analogue of Theorem B: a three-part class of a Green polynomial sees at most *five* of the  $p(n)$  coefficients  $c_{\lambda, \nu}$ , and in general a  $k$ -part class sees at most  $B_k$  of them, the  $k$ -th Bell number.

## 1 The object

Work in  $\Lambda_n$ , the degree- $n$  part of the ring of symmetric functions over  $\mathbb{Q}$ , with the Hall inner product, for which  $\langle p_\rho, p_\sigma \rangle = \delta_{\rho\sigma} z_\rho$  and  $\langle h_\nu, m_\sigma \rangle = \delta_{\nu\sigma}$ . For a partition  $\rho$  write  $\ell(\rho)$  for its number of parts,  $m_i(\rho)$  for the multiplicity of  $i$  among its parts,  $z_\rho = \prod_i i^{m_i(\rho)} m_i(\rho)!$ , and

$$d_\rho := \prod_{i \geq 1} m_i(\rho)! = \frac{z_\rho}{\rho_1 \rho_2 \cdots \rho_{\ell(\rho)}}.$$

Thus  $d_\rho$  is the order of the group of permutations of the *positions* of  $\rho$  that fix the sequence of parts: the automorphism group of  $\rho$  viewed as a labelled multiset.

**Definition 1.**  $M = M(n)$  is the  $p(n) \times p(n)$  matrix with rows and columns indexed by partitions of  $n$  and

$$M_{\rho\nu} = \langle p_\rho, h_\nu \rangle.$$

For  $1 \leq L \leq n$ ,  $M^{(L)}$  is the submatrix on  $\{\rho \vdash n : \ell(\rho) \leq L\} \times \{\nu \vdash n : \ell(\nu) \leq L\}$ .

Two further readings of the same entry will be used throughout. Both are standard; I record them because the proofs below use all three faces of the object.

**Lemma 2.** For all  $\rho, \nu \vdash n$ ,

$$\langle p_\rho, h_\nu \rangle = [m_\nu] p_\rho = \# \left\{ f : [\ell(\rho)] \rightarrow [\ell(\nu)] \mid \sum_{i \in f^{-1}(j)} \rho_i = \nu_j \text{ for all } j \right\}.$$

In particular  $M$  is a matrix of non-negative integers, and  $M_{\rho\nu}$  is the number of ordered set partitions  $(B_1, \dots, B_{\ell(\nu)})$  of the index set of the parts of  $\rho$  with  $\sum_{i \in B_j} \rho_i = \nu_j$ .

*Proof.* The first equality is  $\langle h_\nu, m_\sigma \rangle = \delta_{\nu\sigma}$  applied to the  $m$ -expansion of  $p_\rho$ . For the second,  $p_\rho = \prod_{i=1}^{\ell(\rho)} (\sum_v x_v^{\rho_i})$ , so expanding the product indexes the terms by the choice, for each  $i$ , of a variable  $x_{v_i}$ ; the resulting monomial is  $\prod_v x_v^{s_v}$  with  $s_v = \sum_{i: v_i=v} \rho_i$ . Collecting the terms whose exponent multiset is  $\nu$  and specialising to the monomial  $x_1^{\nu_1} \cdots x_{\ell(\nu)}^{\nu_{\ell(\nu)}}$  identifies the count with the number of maps  $f : i \mapsto v_i$  onto the prescribed exponents, i.e. with the displayed set.  $\square$

**Remark 3.** Every part  $\nu_j$  is  $\geq 1$  and every  $\rho_i$  is  $\geq 1$ , so the condition  $\sum_{i \in f^{-1}(j)} \rho_i = \nu_j$  already forces  $f^{-1}(j) \neq \emptyset$ . The surjectivity of  $f$  is therefore not an extra hypothesis but a consequence — a point that matters in §7, where the obvious control built on it turns out to be vacuous, and in Remark 6, where positivity of the parts of  $\nu$  is shown to be load-bearing for Theorem 4.

## 2 The structure theorem

**Theorem 4** (structure of  $M^{(L)}$ ). Let  $\rho, \nu \vdash n$ .

- (i) If  $\ell(\nu) > \ell(\rho)$  then  $M_{\rho\nu} = 0$ .
- (ii) If  $\ell(\nu) = \ell(\rho)$  and  $\nu \neq \rho$  then  $M_{\rho\nu} = 0$ .
- (iii)  $M_{\rho\rho} = d_\rho = \prod_i m_i(\rho)!$ .

Consequently, for any linear order on partitions of  $n$  refining the order by  $\ell$ , the matrix  $M$  is lower triangular with diagonal  $(d_\rho)_\rho$ ; the same holds for every leading block  $M^{(L)}$ , and  $M^{(L)}$  is exactly the submatrix of  $M$  cut out by  $\ell \leq L$ .

*Proof.* Fix  $\rho, \nu$  and use the third description of Lemma 2. By Remark 3 every such  $f$  is surjective, so  $\ell(\nu) \leq \ell(\rho)$ : this is (i).

Now suppose  $\ell(\nu) = \ell(\rho) = k$ . A surjection between two finite sets of the same cardinality  $k$  is a bijection. So each  $f$  counted is a bijection  $[k] \rightarrow [k]$ , every fibre is a singleton, and the constraint reads  $\rho_i = \nu_{f(i)}$  for all  $i$ . Hence such an  $f$  exists only if  $(\rho_1, \dots, \rho_k)$  and  $(\nu_1, \dots, \nu_k)$  agree as multisets; both being weakly decreasing, that forces  $\nu = \rho$ , which is (ii). When  $\nu = \rho$ , the  $f$  counted are precisely the bijections of  $[k]$  with  $\rho_i = \rho_{f(i)}$  for all  $i$  — the automorphisms of  $\rho$  as a labelled multiset — and there are  $\prod_i m_i(\rho)!$  of them. That is (iii).

The final sentence is immediate: (i) and (ii) say  $M_{\rho\nu} = 0$  unless  $\ell(\nu) < \ell(\rho)$  or  $\nu = \rho$ , so in any order refining  $\ell$  all nonzero off-diagonal entries lie strictly below the diagonal. The set  $\{\ell \leq L\}$  is an initial segment of such an order, so  $M^{(L)}$  is a leading principal submatrix.  $\square$

**Remark 5.** The whole content is the sentence *a surjection between finite sets of equal cardinality is a bijection*. The earlier proof of (i) stopped at “block-triangular”; the step from there to (ii) is free, and it is (ii) that makes the determinant a product and the inverse computable blockwise. This is the shape answer the question asked for: the length filtration on  $\Lambda_n$  does not merely triangularise the  $p$ -vs- $h$  pairing, it *diagonalises* it on each graded piece.

**Remark 6** (positivity of  $\nu$  is load-bearing). Part (ii) fails the moment zero parts are admitted. Extend the column index to weak compositions: for  $\rho = (3, 2)$  and  $\nu = (5, 0)$  both of length 2 and  $\rho \neq \nu$ , the count of maps  $[2] \rightarrow [2]$  with block sums  $(5, 0)$  is 1 (send both parts to block 1), not 0. So (ii) is a statement about partitions, not about matrices of this general shape. This is a fact about the statement, not about the proof.

### 3 The determinant

**Corollary 7.** *For every  $n \geq 1$  and  $1 \leq L \leq n$ ,*

$$\det M^{(L)} = \prod_{\substack{\rho \vdash n \\ \ell(\rho) \leq L}} \prod_{i \geq 1} m_i(\rho)! = \prod_{\substack{\rho \vdash n \\ \ell(\rho) \leq L}} \frac{z_\rho}{\rho_1 \cdots \rho_{\ell(\rho)}}$$

*In particular  $\det M^{(1)} = 1$ ,  $M^{(L)}$  is invertible over  $\mathbb{Q}$  for every  $L$ , and*

$$\det M^{(2)} = 1 + \llbracket n \text{ even} \rrbracket.$$

*Proof.* By Theorem 4 the matrix is triangular with diagonal  $(d_\rho)$ . For  $L = 1$  the only index is  $\rho = (n)$ , with  $d_{(n)} = m_n! = 1$ . For  $L = 2$  the indices are  $(n)$  and the two-part partitions  $(a, b)$ ,  $a \geq b \geq 1$ ;  $d_{(a,b)} = 1$  unless  $a = b$ , when it is  $2! = 2$ , and  $(a, a)$  occurs among partitions of  $n$  exactly when  $n$  is even.  $\square$

**Corollary 8** (when the change of basis is unimodular).  $\det M^{(L)} = 1$  *if and only if every  $\rho \vdash n$  with  $\ell(\rho) \leq L$  is multiplicity-free, if and only if*

$$L = 1, \quad \text{or} \quad n = 1, \quad \text{or} \quad (L = 2 \text{ and } n \text{ odd}).$$

*Proof.*  $\det M^{(L)} = 1$  iff  $d_\rho = 1$  for all  $\rho$  in range, i.e. iff no such  $\rho$  has a repeated part. For  $L = 1$  or  $n = 1$  there is nothing to check. For  $L = 2$ : the only partition of  $n$  into at most two parts with a repeat is  $(n/2, n/2)$ , which exists iff  $n$  is even. For  $L \geq 3$  and  $n \geq 3$  the partition  $(n-2, 1, 1)$  — read as  $(1, 1, 1)$  when  $n = 3$  — has at most three parts and a repeated part; for  $L \geq 2$  and  $n = 2$  the partition  $(1, 1)$  does. So the listed cases are exactly the unimodular ones.  $\square$

**Remark 9.** Taking  $L = n$  gives  $\det M = \prod_{\rho \vdash n} d_\rho$ , the sequence 1, 2, 6, 96, 2880, 9953280, ... for  $n = 1, 2, \dots$ . This has an independent check that uses none of the machinery above. Since  $M(p, s) = M(p, m) M(m, s)$  and the Kostka matrix  $M(s, m)$  is unitriangular for dominance order,  $|\det M(m, s)| = 1$ , so  $|\det M|$  must equal the absolute value of the determinant of the character table of  $S_n$ . It does, for every  $n \leq 7$  (§7, D).

### 4 The inverse, and why the filtration survives it

**Theorem 10** (Möbius form of the inverse). *Let  $\Pi_k$  be the lattice of set partitions of  $[k]$ . For  $\lambda \vdash n$  with  $\ell(\lambda) = k$  and  $\pi \in \Pi_k$  write  $\lambda^\pi$  for the partition of  $n$  whose parts are the block sums  $(\sum_{i \in B} \lambda_i)_{B \in \pi}$ . Then*

$$p_\lambda = \sum_{\pi \in \Pi_k} \tilde{m}_{\lambda^\pi}, \quad \text{and} \quad (M^{-1})_{\lambda\nu} = \frac{1}{d_\lambda} \sum_{\substack{\pi \in \Pi_k \\ \lambda^\pi = \nu}} \prod_{B \in \pi} (-1)^{|B|-1} (|B| - 1)!,$$

where  $\tilde{m}_\mu = d_\mu m_\mu$  is the augmented monomial symmetric function.

*Proof.* Write  $\tilde{m}_{(a_1, \dots, a_r)} = \sum x_{v_1}^{a_1} \cdots x_{v_r}^{a_r}$ , summed over tuples of *distinct* variables; this depends only on the multiset  $\{a_i\}$  and equals  $d_\mu m_\mu$  for  $\mu = \text{sort}(a)$ , since exactly  $d_\mu$  orderings of the exponents give each monomial. Now  $p_\lambda = \sum_{v_1, \dots, v_k} x_{v_1}^{\lambda_1} \cdots x_{v_k}^{\lambda_k}$  over *all* tuples; sorting the tuples by the set partition  $\pi$  of  $[k]$  recording which  $v_i$  coincide, and summing the exponents within each block, gives  $p_\lambda = \sum_{\pi \in \Pi_k} \tilde{m}_{\lambda^\pi}$  as claimed.

This is a sum over the interval  $[\hat{0}, \hat{1}]$  of  $\Pi_k$  in disguise: define  $F(\pi) = \tilde{m}_{\lambda^\pi}$  and  $G(\pi) = \sum_{\sigma \geq \pi} F(\sigma)$ . The displayed identity applied to each block of  $\pi$  separately gives  $G(\pi) = \prod_{B \in \pi} p_{\lambda|_B}$  — the power-sum product over the blocks of  $\pi$  — and  $F(\hat{0}) = \tilde{m}_\lambda$ . Möbius inversion in  $\Pi_k$  therefore yields

$$\tilde{m}_\lambda = \sum_{\pi \in \Pi_k} \mu(\hat{0}, \pi) \prod_{B \in \pi} p_{\lambda|_B} = \sum_{\pi \in \Pi_k} \left( \prod_{B \in \pi} (-1)^{|B|-1} (|B| - 1)! \right) p_{\lambda^\pi},$$

using  $\mu(\hat{0}, \pi) = \prod_{B \in \pi} (-1)^{|B|-1} (|B| - 1)!$  for the partition lattice. Dividing by  $d_\lambda$  expresses  $m_\lambda$  in the  $p$ -basis. Finally,  $M$  is the transition matrix  $p_\lambda = \sum_\nu M_{\lambda\nu} m_\nu$  (Lemma 2), so the matrix expressing  $m$  in terms of  $p$  is exactly  $M^{-1}$ , and reading off the coefficient of  $p_\nu$  gives the formula.  $\square$

**Remark 11.** Theorem 10 is classical — it is the partition-lattice form of the  $m$ -to- $p$  transition, in the spirit of Doubilet’s Möbius-function treatment of the transition matrices. I reprove it here because what I need from it is not the formula but its *support*, and because the normalisation  $1/d_\lambda$  is precisely the quantity Theorem 4(iii) produced from the other side.

**Corollary 12** (the filtration survives inversion).  $(M^{-1})_{\lambda\nu} = 0$  unless  $\ell(\nu) \leq \ell(\lambda)$ , and for every  $L$

$$(M^{(L)})^{-1} = (M^{-1})^{(L)}.$$

*Proof.* A set partition  $\pi \in \Pi_{\ell(\lambda)}$  has at most  $\ell(\lambda)$  blocks, so  $\ell(\lambda^\pi) \leq \ell(\lambda)$  and the sum in Theorem 10 is empty unless  $\ell(\nu) \leq \ell(\lambda)$ . The second statement is the general fact that for a block-triangular  $T = \begin{pmatrix} A & 0 \\ C & D \end{pmatrix}$  one has  $T^{-1} = \begin{pmatrix} A^{-1} & 0 \\ -D^{-1}CA^{-1} & D^{-1} \end{pmatrix}$ , whose leading block is  $A^{-1}$ ; here  $A = M^{(L)}$  because  $\{\ell \leq L\}$  is an initial segment.  $\square$

**Corollary 13** (exact denominators). *Every entry of the row of  $M^{-1}$  indexed by  $\nu$  lies in  $\frac{1}{d_\nu} \mathbb{Z}$ , and the least common denominator of that row is exactly  $d_\nu$ . Hence if  $Y_\rho = \sum_\nu M_{\rho\nu} c_\nu$ , then  $c \mapsto Y$  is integral, while*

$$d_\nu c_\nu \in \mathbb{Z}\text{-span}\{Y_\rho : \ell(\rho) \leq \ell(\nu)\},$$

and  $d_\nu$  cannot be replaced by any proper divisor.

*Proof.* Theorem 10 exhibits each entry of row  $\nu$  as  $1/d_\nu$  times an integer; Theorem 4(iii) and Corollary 12 give the diagonal entry  $(M^{-1})_{\nu\nu} = 1/d_\nu$  exactly, whose denominator is already  $d_\nu$ . The support statement is Corollary 12.  $\square$

**Remark 14.** Corollary 13 is the arithmetic content of the length-filtration duality, and it is sharp in both directions: passing from the  $h$ -coefficients to the  $p$ -class values costs nothing, and passing back costs exactly the automorphism orders of the classes — no primes beyond those dividing  $\prod_k m_k(\nu)!$  ever appear. By Corollary 8 the duality is an isomorphism of *integer* lattices only in the three listed degenerate cases.

## 5 $(M^{(3)})^{-1}$ explicitly

**Theorem 15.** Index rows and columns by  $\{\rho \vdash n : \ell(\rho) \leq 3\}$  and write  $d_\rho = \prod_k m_k(\rho)!$ . Then  $N = (M^{(3)})^{-1}$  is given by

$$N_{\rho\nu} = \begin{cases} 1, & \ell(\rho) = 1, \nu = (n), \\ -\frac{1}{d_\rho}, & \ell(\rho) = 2, \nu = (n), \\ \frac{2}{d_\rho}, & \ell(\rho) = 3, \nu = (n), \\ -\frac{m_{\nu_1}(\rho) + m_{\nu_2}(\rho)}{d_\rho(1 + \llbracket \nu_1 = \nu_2 \rrbracket)}, & \ell(\rho) = 3, \ell(\nu) = 2, \\ \frac{\delta_{\rho\nu}}{d_\rho}, & \ell(\rho) = \ell(\nu) \in \{2, 3\}, \\ 0, & \ell(\nu) > \ell(\rho). \end{cases}$$

*Proof.* Write  $M^{(3)} = \begin{pmatrix} D_1 & 0 & 0 \\ A_{21} & D_2 & 0 \\ A_{31} & A_{32} & D_3 \end{pmatrix}$  in the length grading, so that by Theorem 4 each  $D_k$  is the diagonal matrix  $\text{diag}(d_\rho)_{\ell(\rho)=k}$ . The off-diagonal blocks are read off Lemma 2:

- $A_{k1}$ : for  $\nu = (n)$  there is exactly one map  $[\ell(\rho)] \rightarrow [1]$  and its block sum is  $n$ , so every entry of the first column is 1.
- $A_{32}$ : let  $\ell(\rho) = 3$  and  $\ell(\nu) = 2$ . A surjection  $[3] \rightarrow [2]$  has one singleton fibre and one fibre of size 2; the two cases  $|f^{-1}(1)| = 1$  and  $|f^{-1}(2)| = 1$  are disjoint. In the first,  $f^{-1}(1) = \{i\}$  with  $\rho_i = \nu_1$  (and then the remaining two parts sum to  $n - \nu_1 = \nu_2$  automatically); in the second,  $f^{-1}(2) = \{i\}$  with  $\rho_i = \nu_2$ . Hence  $M_{\rho\nu} = m_{\nu_1}(\rho) + m_{\nu_2}(\rho)$ .

Block inversion of a lower block-triangular matrix gives

$$N = \begin{pmatrix} D_1^{-1} & 0 & 0 \\ -D_2^{-1}A_{21}D_1^{-1} & D_2^{-1} & 0 \\ D_3^{-1}(A_{32}D_2^{-1}A_{21} - A_{31})D_1^{-1} & -D_3^{-1}A_{32}D_2^{-1} & D_3^{-1} \end{pmatrix}.$$

All entries but the  $(3, 1)$  block are now immediate ( $D_1 = (1)$ ,  $A_{21}$  and  $A_{31}$  all ones). For the  $(3, 1)$  block, with  $\ell(\rho) = 3$ ,

$$N_{\rho,(n)} = \frac{1}{d_\rho} \left( \sum_{\ell(\nu)=2} \frac{m_{\nu_1}(\rho) + m_{\nu_2}(\rho)}{d_\nu} - 1 \right), \quad d_\nu = 1 + \llbracket \nu_1 = \nu_2 \rrbracket.$$

Evaluate the sum. Group the two-part partitions  $\nu$  of  $n$  by the unordered pair  $\{\nu_1, \nu_2\} = \{c, n-c\}$ . If  $c \neq n/2$  the pair contributes  $m_c(\rho) + m_{n-c}(\rho)$  with  $d_\nu = 1$ ; the single pair with  $c = n/2$  contributes  $2m_{n/2}(\rho)/2 = m_{n/2}(\rho)$ . Summing over all pairs therefore gives  $\sum_{c=1}^{n-1} m_c(\rho)$ . Since  $\rho$  has three parts, each at most  $n-2$ , every part is counted, so the sum equals  $\ell(\rho) = 3$  and  $N_{\rho,(n)} = (3-1)/d_\rho = 2/d_\rho$ .  $\square$

**Remark 16.** The first column of the inverse continues in the obvious way:  $1, -1/d_\rho, 2/d_\rho$  for  $\ell(\rho) = 1, 2, 3$ . Theorem 10 identifies the pattern, since the only  $\pi$  with  $\lambda^\pi = (n)$  is the one-block partition:

$$(M^{-1})_{\rho,(n)} = \frac{(-1)^{\ell(\rho)-1}(\ell(\rho)-1)!}{d_\rho} = [p_n] m_\rho.$$

At  $\rho = (1^n)$  this is  $(-1)^{n-1}/n$ , which is the classical coefficient  $[p_n]e_n$  — an independent anchor on the formula at the far end of the filtration.

## 6 Theorem B at $L = 3$ , and the Bell bound

Recall the Hall–Littlewood setting of the companion paper:  $Q'_\lambda$  is the modified Hall–Littlewood function,  $Y_\rho^\lambda(t) = \langle p_\rho, Q'_\lambda \rangle$ , and  $Q'_\lambda = \sum_\nu c_{\lambda,\nu}(t)h_\nu$ . Pairing with  $p_\rho$  gives  $Y_\rho^\lambda = \sum_\nu M_{\rho\nu}c_{\lambda,\nu}$ , so every row of  $M$  is a closed-form reduction of a class of Green polynomials to  $h$ -coefficients. The case  $\ell(\rho) = 2$  is Theorem B of that paper. Theorem 4 and the  $A_{32}$  computation give the next case for free.

**Theorem 17** (Theorem B at  $L = 3$ ). *Let  $\rho = (x, y, z) \vdash n$  with  $x \geq y \geq z \geq 1$ . Then for every  $\lambda \vdash n$ ,*

$$Y_{(x,y,z)}^\lambda = c_{\lambda,(n)} + \sum_{i=1}^3 (1 + \mathbb{I}[2\rho_i = n]) c_{\lambda, \text{sort}(\rho_i, n-\rho_i)} + \left( \prod_k m_k(\rho)! \right) c_{\lambda,\rho}.$$

*Proof.* Expand  $Y_\rho^\lambda = \sum_\nu M_{\rho\nu}c_{\lambda,\nu}$  and use Theorem 4: only  $\ell(\nu) \leq 3$  contributes, the term  $\ell(\nu) = 3$  is  $d_\rho c_{\lambda,\rho}$ , and  $\ell(\nu) = 1$  contributes  $c_{\lambda,(n)}$ . For  $\ell(\nu) = 2$  the coefficient is  $m_{\nu_1}(\rho) + m_{\nu_2}(\rho)$  as computed in the proof of Theorem 15. Reindexing that sum by the position  $i$  rather than by  $\nu$ : each  $i$  with  $\rho_i = \nu_1$  or  $\rho_i = \nu_2$  contributes 1, and  $\nu$  is then forced to be  $\text{sort}(\rho_i, n - \rho_i)$ ; the only overcount is at  $\nu_1 = \nu_2 = n/2$ , where a position with  $\rho_i = n/2$  is counted by both  $m_{\nu_1}$  and  $m_{\nu_2}$ , which is the factor  $1 + \mathbb{I}[2\rho_i = n]$ . Note  $1 \leq \rho_i \leq n - 2$ , so  $\text{sort}(\rho_i, n - \rho_i)$  really has two positive parts.  $\square$

The general statement behind both theorems is about the *support* of a row.

**Proposition 18.** *For  $\rho \vdash n$ ,  $M_{\rho\nu} \neq 0$  if and only if  $\nu = \rho^\pi$  for some set partition  $\pi$  of  $[\ell(\rho)]$  — that is, if and only if  $\nu$  is obtained from  $\rho$  by merging parts. Consequently*

$$\#\{\nu : M_{\rho\nu} \neq 0\} \leq B_{\ell(\rho)},$$

*the  $\ell(\rho)$ -th Bell number, and a class of  $\ell(\rho) = k$  parts determines  $Y_\rho^\lambda$  from at most  $B_k$  of the  $p(n)$  coefficients  $c_{\lambda,\nu}$ .*

*Proof.* By Lemma 2 a nonzero entry is witnessed by a surjection  $f : [\ell(\rho)] \rightarrow [\ell(\nu)]$  whose fibres have  $\rho$ -sums  $\nu_j$ ; its fibres form a set partition  $\pi$  with  $\rho^\pi = \nu$ , and conversely any such  $\pi$  gives at least one  $f$ . The bound is  $|\Pi_k| = B_k$ .  $\square$

**Example 19.**  $B_1 = 1$ ,  $B_2 = 2$ ,  $B_3 = 5$ : the one-part class sees only  $c_{\lambda,(n)}$ , the two-part class sees two coefficients (Theorem B), the three-part class at most five (Theorem 17). The bound 5 is attained, e.g. at  $\rho = (3, 2, 1) \vdash 6$ , which sees  $c_{\lambda,\nu}$  for  $\nu \in \{(6), (5, 1), (4, 2), (3, 3), (3, 2, 1)\}$  — five of the eleven coefficients at  $n = 6$ . It is not attained when merges collide:  $\rho = (2, 2, 2) \vdash 6$  sees only three, because all three two-element merges give  $\nu = (4, 2)$ .

## 7 Verification

All computations are exact: integer arithmetic for  $M$ ,  $\mathbb{Q}$  (Python `Fraction`) for determinants and inverses, and `sympy` polynomials in  $t$  for §6. Scripts: `scratch/2026-10-08-c1/{mpm,verify_structure,control}`. SageMath is not installed; everything is pure Python/SymPy, which is the point — the symmetric-function ground truth is reached by polynomial algebra rather than by a combinatorics library, so it is an independent mechanism and not a second reading of one engine.

**Falsifiable-instance census, computed before any agreement was reported.** Claim (ii) of Theorem 4 is the new content, and it is *vacuous* whenever every length class is a singleton. That is exactly  $n \leq 3$ : the degenerate boundary of this problem.

| $n$ | $p(n)$ | (i)-instances | (ii)-instances | non-vacuous (iii) ( $d_\rho > 1$ ) |
|-----|--------|---------------|----------------|------------------------------------|
| 1   | 1      | 0             | <b>0</b>       | 0                                  |
| 2   | 2      | 1             | <b>0</b>       | 1                                  |
| 3   | 3      | 3             | <b>0</b>       | 1                                  |
| 4   | 5      | 9             | 2              | 3                                  |
| 5   | 7      | 19            | 4              | 4                                  |
| 6   | 11     | 48            | 14             | 7                                  |
| 7   | 15     | 92            | 26             | 10                                 |
| 8   | 22     | 201           | 60             | 16                                 |
| 9   | 30     | 379           | 112            | 22                                 |

**A. Two engines for the entry.** Engine A counts the maps of Lemma 2 (dynamic programming over the parts, itself checked against brute-force enumeration of all  $\ell(\nu)^{\ell(\rho)}$  maps: 209 pairs, 0 mismatches). Engine B multiplies out  $p_\rho = \prod_i (\sum_v x_v^{\rho_i})$  as a polynomial and reads the coefficient of  $x_1^{\nu_1} \cdots x_k^{\nu_k}$ . Over all  $n \leq 9$ : **1818 pairs compared, 736 nonzero, 0 mismatches**.

**B. The structure theorem.** Direct check of (i)/(ii)/(iii) for all  $n \leq 9$ : 752 instances of (i), 218 instances of (ii), 64 non-vacuous instances of (iii) — **0 failures**.

**C. The determinant.** Corollary 7 against exact Gaussian elimination, for *every*  $(n, L)$  with  $n \leq 9$ : **0 mismatches**.  $L = 1$  returns 1 in every case, the free sanity check. Corollary 8's characterisation of  $\det = 1$ : all  $(n, L)$  with  $n \leq 10$ , 0 mismatches.

**D. The determinant, through a different matrix.**  $|\det(\text{character table of } S_n)|$  computed by Murnaghan–Nakayama (validated by column orthogonality  $\sum_\rho (\chi_\rho^\lambda)^2 / z_\rho = 1$  for every  $\lambda$ ) equals  $\prod_{\rho \vdash n} d_\rho$  for  $n \leq 7$ : 1, 2, 6, 96, 2880, 9953280, 100329062400.

**E. The inverse.** Theorem 10 against Gaussian elimination: 434 entries,  $n \leq 7$ , 204 nonzero, 0 mismatches. Corollary 12:  $(M^{(L)})^{-1}$  versus the truncation of  $M^{-1}$ , all  $(n, L)$  with  $n \leq 7$ , 0 mismatches. Corollary 13: for every row  $\nu$  (44 rows,  $n \leq 7$ ) the l.c.m. of the denominators is exactly  $d_\nu$ . Theorem 15 against Gaussian elimination: **608 entries,  $n \leq 10$ , 192 nonzero, 0 mismatches**.

**F. Theorem 17, through a chain that never counts a set partition.**  $Q'_\lambda = \sum_\nu K_{\nu\lambda}(t)s_\nu$  with  $K$  from the *charge statistic* on semistandard tableaux;  $Y_\rho^\lambda = \sum_\nu K_{\nu\lambda}(t)\chi_\rho^\nu$  with  $\chi$  from Murnaghan–Nakayama;  $c_{\lambda,\nu} = \sum_\mu K_{\mu\lambda}(t)[h_\nu]s_\mu$  with  $[h_\nu]s_\mu$  from the Jacobi–Trudi determinant (validated by  $M(s, h)M(h, s) = I$  for  $n \leq 7$ ). No step uses  $M_{\rho\nu}$ . Result: **115 checks,  $3 \leq n \leq 7$ , all with  $Y_\rho^\lambda \neq 0$ , 0 failures**. Proposition 18: support = coarsenings, all 66 rows with  $n \leq 8$ , 0 mismatches.

**G. Controls.** Each claim gets a planted control, and each silence is diagnosed.

| control                                                        | reading                                                                     | note                              |
|----------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------|
| C0 drop “fibres nonempty”                                      | <b>silent</b> , 0/434                                                       | <i>Vacuous</i> , and mark 3 the g |
| C1 force blocks to be intervals                                | <b>fires</b> 139/434                                                        | 1, so flipping entry. Repla       |
| C2 $d_\rho \rightsquigarrow z_\rho$ in Cor. 7                  | <b>fires</b> 27/28                                                          | smallest wit                      |
| C3 $m_i! \rightsquigarrow m_i$                                 | <b>fires</b> 17/36                                                          | vs 2.                             |
| C4a plant one nonzero in a same-length off-diagonal slot       | checker <b>fires</b> (1 failure); det <b>silent</b> ( $24 \rightarrow 24$ ) | first at $(n, L$                  |
| C4b plant a symmetric pair                                     | <b>fires</b> det $24 \rightarrow 12$                                        | could only f                      |
| C5a Möbius sign flipped                                        | <b>fires</b> 95/434                                                         | census says                       |
| C5b $( B  - 1)!$ dropped                                       | <b>fires</b> 83/434                                                         | fires at all 1                    |
| C6 transpose, truncation vs inversion                          | <b>silent</b>                                                               | <i>Vacuous for</i>                |
| C6' plant one cross-length entry above the length diagonal     | <b>fires</b>                                                                | <i>provably so</i>                |
| C6'' character table, truncate by $\ell(\lambda)$              | <b>fires</b> 243/368 at $n = 6$                                             | perturbation                      |
| C7 admit zero parts in $\nu$                                   | <b>fires</b>                                                                | leaves it tri                     |
| C8 drop $(1 + \llbracket \nu_1 = \nu_2 \rrbracket)$ in Thm. 15 | <b>fires</b> 6/116                                                          | changed. Th                       |
|                                                                |                                                                             | <b>the determ</b>                 |
|                                                                |                                                                             | <b>C is eviden</b>                |
|                                                                |                                                                             | <b>ditional on</b>                |
|                                                                |                                                                             | independent                       |
|                                                                |                                                                             | the control t                     |
|                                                                |                                                                             | first at $\lambda =$              |
|                                                                |                                                                             | of size 3 fir                     |
|                                                                |                                                                             | <i>Vacuous, a</i>                 |
|                                                                |                                                                             | ing principa                      |
|                                                                |                                                                             | with inversi                      |
|                                                                |                                                                             | matrix, upp                       |
|                                                                |                                                                             | is upper tri                      |
|                                                                |                                                                             | C6'/C6''.                         |
|                                                                |                                                                             | two of thre                       |
|                                                                |                                                                             | matrix <i>sing</i>                |
|                                                                |                                                                             | stay invert                       |
|                                                                |                                                                             | blocks inver                      |
|                                                                |                                                                             | ity; the thir                     |
|                                                                |                                                                             | cation/inver                      |
|                                                                |                                                                             | the commut                        |
|                                                                |                                                                             | is not a taut                     |
|                                                                |                                                                             | Remark 6:                         |
|                                                                |                                                                             | ment.                             |
|                                                                |                                                                             | census says                       |
|                                                                |                                                                             | $(n \text{ even}, m_n$            |
|                                                                |                                                                             | 6. First at                       |
|                                                                |                                                                             | $\nu = (2, 2)$ .                  |

**An instrument fault, recorded because it read as a refutation.** The first run of F reported **86 of 115 failures**. The polynomials were equal; the comparison was not. In the Murnaghan–Nakayama recursion the sign was coded as  $(-1)**ht$  with  $ht$  the signed displacement of a  $\beta$ -number, which is *negative*; in Python  $(-1)**(-1)$  is the float  $-1.0$ , so  $\chi_\rho^\nu$  was returned as a

float,  $Y$  carried floats, and `sympy`'s `!=` separated  $t^3+t^2+t-1.0$  from  $t^3+t^2+t-1$ . Every value was correct and every comparison was wrong. Had I read that output as data, I would have recorded a true theorem as refuted at  $n = 4$ . The repair is `1 if ht % 2 == 0 else -1`, after which the character engine passes column orthogonality at  $n = 5, 6$  and  $F$  is clean. *A disagreement between two representations of one value is not a disagreement about the value* — and a 0 failure count is only meaningful once the comparison operator has been shown to compare what it claims to.

## 8 What is and is not new

The  $m$ -to- $p$  transition (Theorem 10) is classical, as is the evaluation of  $\langle p_\rho, h_\nu \rangle$  as an ordered-set-partition count (Lemma 2, which is Lemma 2.3 of the companion paper in the two-part case). The determinant of the  $S_n$  character table is also classical, and by the remark after Corollary 8 it is equivalent to the  $L = n$  case of Corollary 7. I have not searched the literature this session (no browsing), so I claim no novelty for any of those three.

What this session adds is the *filtered* picture and its consequences: Theorem 4(ii) — the diagonal blocks are diagonal, not merely invertible — and with it the closed form for  $\det M^{(L)}$  at every truncation (Corollary 7), the exact unimodularity criterion (Corollary 8), the exact denominators of the duality (Corollary 13), the explicit  $(M^{(3)})^{-1}$  (Theorem 15), the three-part analogue of Theorem B (Theorem 17), and the Bell-number bound on how many  $h$ -coefficients a  $k$ -part class can see (Proposition 18).

**Gaps.** None in the proofs above; every step is an identity, a definition, or Möbius inversion in  $\Pi_k$ . Three things are *not* done here and should not be read as done:

1. No closed form for  $c_{\lambda, \nu}$  at  $\ell(\nu) = 3$  is proved. Theorem 17 reduces the three-part Green polynomial to five  $h$ -coefficients; it does not evaluate them. The missing input is a three-point analogue of Rick's two-point shuffle formula, and whether  $Sh_{A,B}$  composes to a triple shuffle is untouched by this session.
2. Nothing here is formalised in Lean.
3. The claim that Lemma 2 and Theorem 10 are classical is a recollection, not a literature check; no browsing was done.